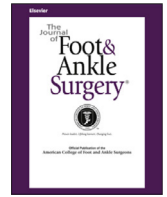




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Comparison of Clinical Outcomes With Arthroscopic ATFL Repair With the CFL Repair and Open ATFL and CFL Repair in Chronic Lateral Ankle Instability

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ABSTRACT

It is important to eliminate instability related to chronic lateral ankle instability (CLAI) to prevent osteoarthritis progression. We performed arthroscopic anterior talofibular ligament (ATFL) repair and performed calcaneofibular ligament (CFL) repair if instability remained. This study aimed to assess the clinical outcomes of our arthroscopic procedure compared to that of open surgery. Forty ankles underwent arthroscopic surgery and 23 ankles underwent open surgery to repair the lateral ankle ligaments for CLAI. In the arthroscopic surgery, varus stress was applied under fluoroscopy after ATFL repair, and CFL repair was performed if instability remained. Open surgery was performed using the Broström procedure with ATFL and CFL repair. To assess clinical outcomes, American Orthopaedic Foot & Ankle Society (AOFAS) and Karlsson-Peterson (KP) scores were collected preoperatively and at the final follow-up. The talar tilt angle (TTA) was measured preoperatively and 1 year postoperatively. The arthroscopic group showed significantly higher AOFAS and KP scores at the final follow-up compared to the open surgery group. There was no significant difference in TTA at 1 year between the groups. In open surgery, 2 patients required revision surgery. There were no major complications, but scar-related pain in 2 cases of open surgery was reported. Arthroscopic ATFL repair with the CFL repair gave satisfactory clinical outcomes compared to open surgery in CLAI because of low invasive to soft tissue including the joint capsule. It is important to minimize soft tissue dissection in repairing the lateral ankle ligament in patients with CLAI.

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Ankle sprains are frequently encountered in daily living and sports activities. More than 20% of ankle sprains develop chronic lateral ankle instability (CLAI) and required surgical treatment to obtain stability of the ankle joint (1). For the repair of the lateral ankle ligaments, the Broström procedure has been performed and satisfactory clinical results have been reported (2). It has been reported that the Broström procedure without calcaneofibular ligament (CFL) repair has obtained excellent outcomes (3), but the classic Broström procedure is defined as the repair of the anterior talofibular ligament (ATFL) and CFL (4,5). Theoretically, repair of both ATFL and CFL will increase stability in cases with both ATFL and CFL instability. A previous report by Maffulli et al showed that 30% of patients who underwent the Broström procedure without CFL repair for CLAI had new radiographic

signs of degenerative changes despite good clinical results (6). Broström reported that an ATFL tear was observed in two-thirds of the ankles operated on, and combined ATFL and CFL rupture was observed in one-quarter in the surgical findings (7). Other reports using oblique CFL view on MRI showed a high rate of CFL injury in patients with CLAI (8). These studies indicate that ATFL repair alone may not be sufficient to stabilize the ankle joint in the case of both ATFL and CFL injuries.

Recently, the arthroscopic Broström procedure has been widely performed as less invasive surgery, and good clinical outcomes have been reported (9–11). There are several reports comparing the arthroscopic and open procedures which showed the superiority of arthroscopic surgery (12,13). On the other hand, another report demonstrated that there was no significant difference in the VAS score, clinical scores, and stress radiography (14). Since these comparative studies included only arthroscopic ATFL repair, the presence or absence of CFL deficiency in CLAI was not taken into account. The purpose of this study was to compare the clinical outcomes of open Broström with ATFL and CFL repair and arthroscopic Broström in addition to CFL repair in cases where CFL deficiency was observed after ATFL repair in patients with CLAI. We

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hypothesized that both procedures could obtain good clinical outcomes but arthroscopic procedures would be superior to open surgery with low complications by less invasion of soft tissues.

Patients and Methods

Participants

Forty ankles from 39 patients who were surgically treated via arthroscopic repair of the lateral ankle ligaments for CLAI between October 2018 and December 2020 were included in this study. The participants consisted of 20 men and 19 women. In 1 patient, both ankles required surgery. Open surgery was performed on 23 ankles in 22 patients with CLAI between November 2007 and August 2018. They consisted of 7 men and 15 women. In 1 patient, both ankles required surgery. All patients included in this study were followed up for more than 1 year after surgery. Patients with an osteochondral lesion of the talus (OLT) or cartilage injury in the talus or tibia, systemic diseases, such as rheumatoid arthritis, revision surgery, large subfibular ossicle more than 5 mm, syndesmosis injury, and deltoid ligament tear with obvious medial instability were excluded from this study.

Evaluation

Clinical outcomes were evaluated using the American Orthopedic Foot and Ankle Society (AOFAS) hindfoot-ankle score and the Karlsson-Peterson (K-P) score, before and after surgery (15–17). For the assessment of preoperative activity, the ankle activity score (AAS) was used, which consists of 53 sports, 3 working activities, and 4 general activities (18). For the radiographic assessment, varus stress radiography was performed with a force of 50 N using a tension device (Imada, Toyohashi, Japan) preoperatively and 1 year postoperatively. Prior to ligament repair, the quality of the ATFL remnants was evaluated arthroscopically according to a previous report about the grading of the remnant quality (19). The remnants of the ATFL were divided into 3 groups: (1) poor, arthroscopic observation of a clearly hypoplastic ligament with poorly defined margins, (2) moderate, arthroscopic observation of fibrotic tissue or synovitis demonstrating a stretched hyperplastic or hypoplastic ligament, and (3) excellent, arthroscopic observation of normal synovial tissue demonstrating a ligament with sharply defined margins. This study was approved by the Ethics Committee of our university, and informed consent was obtained from all individual participants included in this study.

Arthroscopic Surgery

Patients were placed in the supine position, and standard anterolateral (AL) and anteromedial (AM) portals were placed under joint distraction using an ankle distractor (Smith & Nephew, Memphis, TN) and a 30° oblique arthroscope. Then, the accessory AL portal (AAL) was placed as the working portal along the anterior border of the fibula at the level of the superior fascicle of the ATFL. Viewing from the anterolateral portal, a 23 gauge hollow needle with a 3–0 nylon thread was penetrated into the proximal part of the ATFL remnant through the AAL portal (Fig. 1A, B). The loop of the nylon thread was retrieved, and the needle was withdrawn. No. 2 thread (Ultra-braid, Smith & Nephew) was passed through the loop of nylon and sewn using a racking hitch. Then, the thread was placed in the ATFL remnant 2 or 3 additional times using a 23 gauge needle (Fig. 1C). If the quality of the remnant was poor, suture tape (Ultra-tape, Smith & Nephew) was used instead of No. 2 thread to avoid cutting off the remnant. Thereafter, a drill hole was created in the footprint of the superior fascicle of the ATFL via the AAL portal. A knotless anchor (Bioruptor, Smith & Nephew) with the No. 2 thread or suture tape was inserted into the fibula at the 0° position of the ankle joint (Fig. 1D). Subsequently, varus stress with a force of approximately 50 N was provided to the ankle joint manually, and talar tilting angle was confirmed under fluoroscopy. If the talar tilting angle remains, CFL repair was performed through the AAL portal. The ALL portal was extended toward the fibula tip with a total length of 15 mm. Peroneal tendons were retracted laterally and CFL was confirmed. A soft anchor (Q-Fix mini, Smith & Nephew) was inserted into the footprint of the CFL. One end of the anchor was passed into the CFL and tied using a sliding knot technique under direct visualization (Fig. 1E) (20). Finally, varus stress was performed again to confirm the absence of instability. After surgery, the ankle joint was fixed with an ankle brace with cast stays on both sides (Fit cure-ankle, ALCARE Co., Ltd, Tokyo, Japan) for 2 weeks. Weightbearing was permitted the day after surgery. Jogging was permitted at 6 weeks, and return to sports activity was allowed at 3 months. All arthroscopic surgeries were performed by 2 surgeons who experienced more than 20 and 13 years of orthopedics (T.N. and Y.I.).

Open Surgery

Anteromedial and anterolateral portals were placed, and intra-articular pathological lesions including ATFL were evaluated. After diagnostic arthroscopy, a skin incision of 5 cm in length was made along the lateral malleolus to the talar attachment of the ATFL. The capsule and periosteum at the distal fibula were incised along with the ATFL (Fig. 2A). CFL was identified by the retraction of the peroneal tendon. The fibula attachments of the ATFL and CFL were exposed, and decortication was performed. Looped No. 1 braided nylon thread (Surgilon™ Braided Nylon Sutures, Covidien, Mansfield, MA) was placed

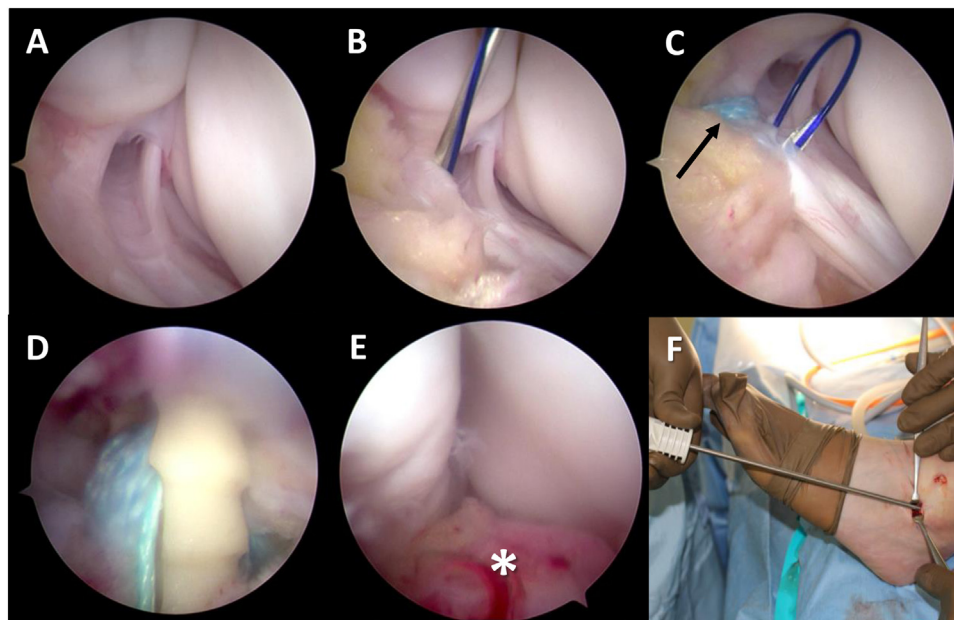


Fig. 1. Procedure for arthroscopic anterior talofibular ligament (ATFL) repair. (A) An ATFL remnant was observed from the anterolateral portal. (B) The nylon thread was set at the proximal one-third of the remnant using a 23G hollow needle, and then suture tape was placed in the ATFL using the loop of the nylon thread. (C) The thread was placed in the ATFL remnant 2 or 3 additional times using a 23 gauge needle. Arrow indicates suture tape placed in the ATFL. (D) The knotless anchor with suture tape was inserted to the attachment of the fibula. (E) After the insertion of the knotless anchor. *, the repaired ATFL. (F) Calcaneofibular ligament (CFL) repair through the accessory portal.

into the proximal third of the ATFL and CFL. Two or 3 threads were used for ATFL, and 1 or 2 threads were for CFL. Five or 6 bone tunnels were created at the fibula attachment by using a 1.2 mm K-wire, and No. 1 threads of ATFL and CFL were pulled out (Fig. 2B, C). The threads were pulled at manual maximum power and tied strongly to the fibula at 0° position of the ankle joint (Fig. 2D). Subsequently, the capsule was repaired using a 3-0 absorbable thread (Vicril, Ethicon, Summerville, NJ), and the skin was closed. After surgery, a lower leg cast or hard brace was applied for 3 weeks, after which ROM excise and partial weightbearing were started. Full weightbearing was permitted at 6 weeks, and return to sports activity was allowed at 3 months. All open surgeries were performed by 2 surgeons who experienced more than 30 and 20 years of orthopedics (N.A. and T.N.).

Statistical Analysis

A post hoc power analysis of a 2-tailed independent *t* test was performed using G*Power 3.1.9.7 for Windows (Heine University, Düsseldorf, Germany). The effect size used for this assessment was large ($d = 0.8$), and the alpha level used was $p < .05$. The sample size of 40 in the arthroscopic group and 23 in open surgery groups were used. The post hoc analysis revealed the statistical power for this study was 0.852. Thus, there was adequate power at the large effect size level. The effect size from Cohen's *d* was employed to compare the dependent variables of 2 groups such as AOFAS score, K-P score and TTA in the present study. Significant differences between the 2 groups were analyzed using the Mann-Whitney *U* test. Statistical significance was defined at the 5% ($p \leq .05$) level.

Results

The age in the arthroscopic surgery group (33.8 ± 1.8 years) was significantly higher than that in the open surgery (25.4 ± 13.1 years)

($p < .01$), and the follow-up period in the arthroscopic surgery group (16.2 ± 6.8 months, range 12–36 months) was significantly lower than that in the open surgery group (21.5 ± 10.2 , range 12–47 months) ($p < .01$). The mean preoperative AAS in the arthroscopic group was 5.3 ± 1.9 points and that in the open group was 4.9 ± 1.9 points. There was no significant difference in AAS between the 2 groups. Also, no significant difference was found in the preoperative AOFAS score, K-P scores and TTA in both groups. The AOFAS and K-P scores significantly improved from the preoperative to the final follow-up in both groups. However, the AOFAS and K-P scores in the arthroscopic group were significantly higher than those in the surgery open group ($p < .05$) at the final follow-up. TTAs in both groups also significantly improved 1 year after surgery ($p < .05$). There was no difference in TTA at 1 year between the 2 groups (Table).

In the open surgery group, 2 patients had severe ankle sprains and underwent revision surgery by reconstruction using the gracilis tendon due to recurrent instability. One patient complained of ankle instability and hindfoot pain 1 year after surgery and underwent debridement using hindfoot endoscopy. At the final follow-up, 4 patients complained of occasional ankle pain and giving way sensation, and the other 4 patients had occasional ankle pain. In the arthroscopic surgery group, 2 patients had occasional ankle pain and 1 patient, who had a severe ankle sprain 8 months after surgery, complained of ankle instability while walking on uneven ground. With regards to complications, 1

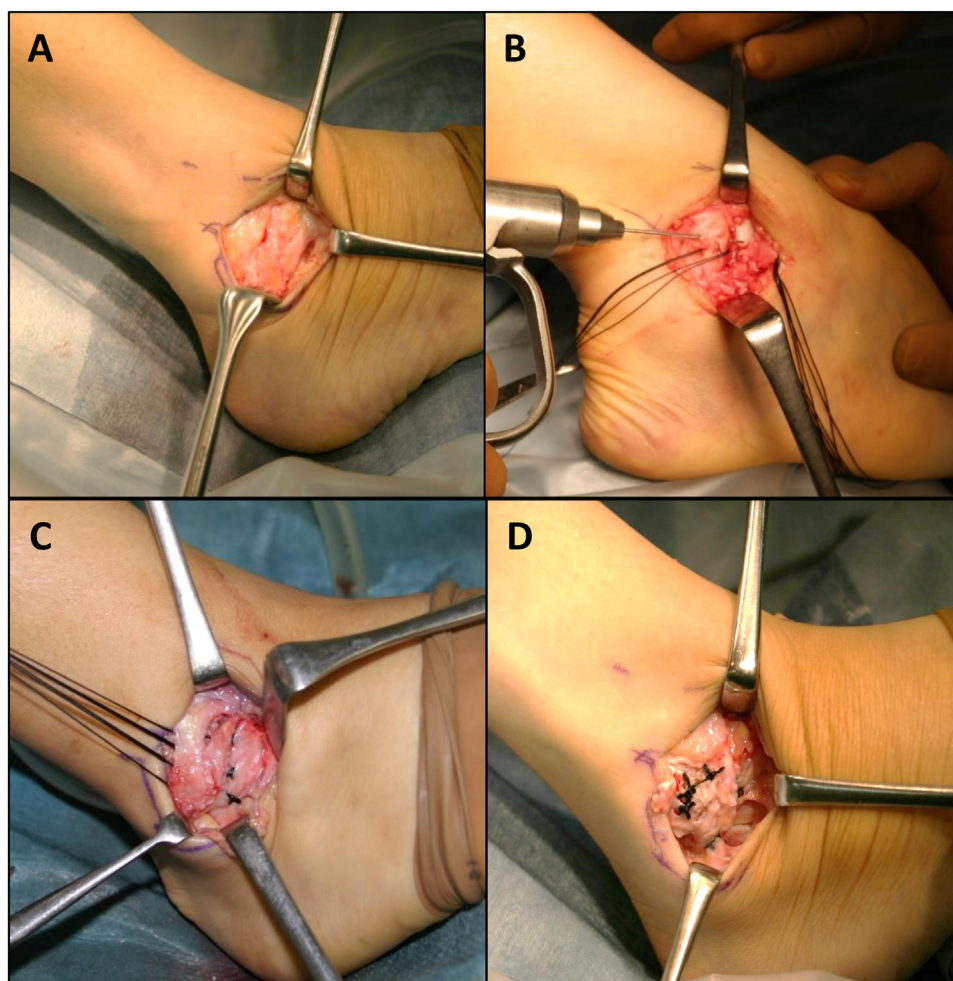


Fig. 2. Open surgery according to the Broström procedure. (A) ATFL and CFL remnants were exposed through the 5 cm length of the skin incision. (B) No. 1 braided nylon threads were tied at the proximal third of the ATFL and CFL remnants, and bone tunnels were created at the fibula. (C) The threads were then pulled out to the fibula attachment. (D) Finally, the threads were securely tied at the fibula.

Table

Comparison of arthroscopic surgery with open surgery groups in the American Orthopedic Foot and Ankle Society (AOFAS) hindfoot-ankle score and the Karlsson-Peterson (K-P) score, and talar tilting angle (TTA)

		Open Surgery (n = 23)	Arthroscopic Surgery (n = 40)	p Value
Preoperative	AOFAS score	71.1 ± 2.7 (66-74)	70.7 ± 4.5 (49-74)	.757
	KP score	66.3 ± 5.2 (58-70)	66.4 ± 5.2 (58-70)	.889
	TTA (°)	11.9 ± 4.9 (3-21)	10.7 ± 4.2 (5-24)	.147
Postoperative	AOFAS score	91.7 ± 8.0 (82-100)	96.4 ± 5.9 (82-100)	.031*
	KP score	91.2 ± 8.8 (70-100)	95.8 ± 6.1 (80-100)	.026*
	TTA (°)	5.7 ± 3.4 (1-14)	4.1 ± 3.0 (0-17)	.139

* There is significant difference between the 2 groups.

patient complained of knot irritation at the distal end of the fibula which was used for the CFL repair in the arthroscopic surgery group. In the open surgery group, 2 patients complained of scar irritation, especially pain from shoes rubbing against the scar during sports. There were no neurological complications in either of the groups.

ATFL remnants were observed arthroscopically in all cases. ATFL remnant in the open group showed 10 (44%) ankles for excellent, 9 (39%) ankles for moderate, and 4 ankles (17%) for poor. In the arthroscopic group, ATFL remnants were 16 (40%) ankles for excellent, 14 (35%) ankles for moderate, and 10 (25%) ankles for poor. Preoperative ATFL remnant quality in both groups was similar.

Discussion

In this study, clinical outcomes between arthroscopic and open Broström procedures were compared, and the AOFAS score at the final follow-up in the arthroscopic Broström procedures was superior to that in the open Broström procedures and subjectively more patients in the open Broström procedures complained of giving way sensation compared to the arthroscopic Broström procedures. In addition, 3 patients needed to have surgery caused of the recurrent instability in the open Broström procedure. Unique features of this study are that the ATFL and CFL dysfunction were absent just after surgery and there was no difference in the ATFL remnant condition at the time of surgery. Thus, there was no effect of the ATFL remnant condition and residual CFL dysfunction on the results in comparison with open and arthroscopic procedures in this study.

One of the major differences between these methods is the degree of resection for the joint capsule. While arthroscopic surgery can preserve the joint capsule, open surgery is required to resect the joint capsule to expose and isolate the ATFL and CFL. A previous cadaveric study showed that the inferior fascicle of ATFL is integrated with the inferior-lateral articular capsule and reinforces it (21). Therefore, the mechanical strength of the anterolateral aspect of the ankle by the joint capsule may be weakened by the open Broström procedure. Blood supply to the ATFL and CFL should be considered. It is reported that there are 4 separate sources of extraosseous blood supply to the lateral ankle ligaments and some of the vascular supplies may be damaged by an ankle sprain (22). A surgical approach to the ATFL and CFL by open Broström procedures may damage the remaining blood supplies, which affects the ligament healing after surgery. Open Broström procedures had more patients complaining of the giving way sensation than those in the arthroscopic Broström procedures, which suggests that the recovery of the functional instability is inadequate. The lateral ankle ligament and joint capsule had many mechanoreceptors which play a crucial role in proprioception, and patients with CLAI exhibited a deficit of proprioception (23-25). Resection of the joint capsule in the open Broström procedures may induce damage of the sensory nerve distributing the

ATFL and CFL, which is a distinct disadvantage to recovering the functional instability.

In the excellent remnant, ATFL was detached from the fibula, which caused ankle instability. Therefore, reattachment of the ATFL remnant enables improved ankle instability in both groups. However, the poor remnant does not have enough strength to stabilize the ankle joint after the reattachment of the remnant to the fibula, which may cause the recurrence of the instability in open surgery. On the other hand, a poor ATFL remnant that is integrated with the joint capsule is reattached to the fibula along with the joint capsule during the arthroscopic procedure, which may obtain more resistant strength against instability compared to the remnant only. Thus, the arthroscopic procedure had less recurrence of instability than the open surgery, although the percentage of the poor remnant in arthroscopic surgery was higher than that in open surgery. Arthroscopic surgery may be more advantageous than open surgery in the CLAI with poor remnants (26).

The superiority of the arthroscopic Broström procedure compared to the open Broström procedure was recently reported (12,13). The meta-analyses revealed higher clinical scores, such as the AOFAS score and lower VAS, in arthroscopic repair than in open surgery (13). Regarding postoperative clinical scores, several studies showed that there was no significant difference between arthroscopic and open surgeries (27-29). Even if there is no significant difference in clinical scores, the arthroscopic procedure has several advantages. Matsui et al showed that the arthroscopic procedure allows an early return to daily activities compared to the open procedure (28). Hou et al demonstrated that the arthroscopic modified Broström procedure showed a significantly shorter period of return to the preinjury sport, although no significant difference was found in clinical scores between the arthroscopic and open modified Broström procedure (30). Minimally invasive surgery for CLAI allows for good outcomes in the return to sport by the preservation of the soft tissue of the lateral aspect of the ankle (31). In particular, arthroscopic repair has a lower scar complication rate, which is similar to the results of our study (14,32-35).

Our study showed that AOFAS and K-P scores at the final follow-up in the arthroscopic procedure were higher than those in the open procedure. In our series, the pre- and postoperative AOFAS scores may be slightly higher than in previous studies because the ankles with the intra-articular lesion such as the osteochondral lesion were excluded (12,13,27-29). Therefore, postoperative scores were essentially affected by the intervention of the lateral ankle ligament. The instability induced by the deficiency of the ATFL increases the stress distribution on the medial gutter, leading to subchondral bone sclerosis (19). The neuropeptides secreted from the sensory nerve have roles not only in pain but also enhancement of the differentiation of the osteoblasts, which results in the sclerotic change of the subchondral bone (36). Therefore, enhancement of bone formation due to the increased stress distribution by the instability may cause pain. At the final follow-up, patients who complained of pain and instability in the open procedure were more frequent than those in the arthroscopic procedure. It is reported that the anterolateral joint capsule contributes to mechanical stability for anterior displacement and tilting angulation of the talus (37). The lateral ankle ligaments and anterolateral joint capsule in open surgery may become dysfunctional over time due to the greater invasion of the joint capsule than in arthroscopic surgery. It is thought that pain induced by the instability and scar may affect the difference in the clinical scores at the final follow-up between the arthroscopic and open procedures. In open surgery, it is important to minimize the dissection of the soft tissue including the joint capsule by creating a small incision just enough to thread and insert the anchor after confirming the running of the ATFL.

This study has several limitations. First, this was a retrospective study. In this study, we compared the clinical outcomes of the arthroscopic and open procedures, and the statistical power was sufficient for a retrospective study. However, these 2 procedures were not performed at the same time and the procedure was not randomly selected because we changed the surgical procedure from open to arthroscopic surgery with the development of the arthroscopic technique in worldwide trends over time. Thus, clinical results in both groups could not be compared prospectively. Prospective studies are required to conclude if one technique is truly superior to another. This is a major limitation of this study. Second, postoperative rehabilitation protocols were different between the 2 groups. Permission to partial weightbearing and removal of the lower leg cast or hard brace were slower in the open surgery than that in the arthroscopic surgery to prevent pain and swelling due to high invasion by open surgery. Third, the anchor system was not used in open surgery. However, as good tension to the ATFL and CFL was provided by the pull-put technique in open surgery, we believe the clinical results in both groups were comparable. Finally, age and follow-up duration were significantly different between the 2 groups. Although age was higher in the arthroscopic group than in the open surgery group, the preoperative AAS and TTA were not significantly different. The follow-up duration in the arthroscopic group was shorter than that in the open surgery group. There is a possibility that longer follow-up periods will reduce the clinical outcomes of arthroscopic surgery. Further investigation is required over a longer follow-up period.

In conclusion, our study showed that arthroscopic Broström procedures had higher postoperative clinical scores and lower complication rates than open Broström procedures, although further prospective studies are needed. It is important to minimize the soft tissue dissection such as the joint capsule in repairing the lateral ankle ligament in patients with CLAI.

Ethical Approval

The study was approved by the local ethics committee of our university.

Informed Consent

Informed consent was obtained from all individual participants.

Authors' Contributions

T.N. collected and analyzed the data and drafted the manuscript. Y.I., J.S., A.N., and S.K. collected and analyzed the data. N.A. gave final approval to the manuscript.

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